

AFCS Smart Campus

Investment Pitching Document

Version 2.0 — July 2026

Air Force Comprehensive School, Igbara-Oke, Ondo State, Nigeria

Production Since: Early 2026

Executive Summary

AFCS Smart Campus is a comprehensive, AI-powered school management operating system purpose-built for Nigerian secondary schools. Developed and deployed at **Air Force Comprehensive School, Igbara-Oke (AFCS)**, it replaces fragmented paper-based processes with an integrated digital platform covering attendance, duty rosters, AI timetable generation, muster parades, daily reporting, prefect management, and multi-channel communications (Telegram, WhatsApp, SMS).

80+

STAFF SERVED

3,000+

STUDENTS TRACKED

50+

API ENDPOINTS

13

AUTOMATION RULES

The system processes **hundreds of daily check-ins** via QR code and manual entry, runs **13 scheduled automation rules** (duty notifications, check-in reminders, absentee alerts, period reminders, daily summaries), and delivers **multi-channel notifications** via Telegram bot (25+ commands), WhatsApp Cloud API, and SMS. An **AI-powered timetable generator** uses constraint-satisfaction algorithms to produce conflict-free timetables. A **global search bar** provides instant access to all platform features.

AFCS Smart Campus represents a **Nigerian-built, Nigerian-tested solution** that addresses the acute need for digital transformation in Africa's educational sector — now with **18 prefect roles, Ollama AI provider support**, and **all automation notifications redirected to Telegram** for reliability and zero-cost delivery.

The Ask: NGN 50,000,000 (~\$32,000) investment to productize, scale, and deploy AFCS Smart Campus to 100+ Nigerian schools within 24 months.

The Problem

Nigerian schools face critical operational challenges:

- **Paper-based record keeping** — Attendance, duties, and reports are handwritten, lost easily, and nearly impossible to audit. A typical Nigerian school processes 500+ attendance records daily on paper.
- **No real-time visibility** — School administrators (commandants, principals) lack live data on who is present, who is on duty, or what tasks are pending — leading to accountability gaps.
- **Manual timetable generation** — Creating conflict-free timetables for 30+ teachers across multiple classes takes 3-5 days of manual work and frequently contains room/teacher/subject clashes.
- **Fragmented communication** — WhatsApp groups, physical notice boards, word-of-mouth. No unified system for task assignment, acknowledgment, or follow-up.
- **No accountability framework** — Task assignment and follow-up are informal. There is no digital audit trail for duty assignments, parade briefings, or attendance overrides.
- **Limited oversight** — Proprietors, education boards, and ministry officials cannot access school performance data remotely. No dashboards for attendance trends or staff performance.
- **No student leadership management** — Prefect assignments are manual, inconsistently tracked, and not integrated with the attendance or behavior monitoring system.

Market Gap

Existing solutions (Edusuite, SchoolTry, SkoolMedia, Eduplana) focus on **academic records** — grades, transcripts, fees, and learning management. None provide integrated **school operations management**: the daily workflow of running a boarding school — attendance tracking, duty rotations, parade management, real-time staff communication, automated reporting, and prefect management. AFCS Smart Campus uniquely fills this gap as the only platform combining operations management + academic scheduling + multi-channel communications + automation.

The Solution

AFCS Smart Campus is a mobile-first, cloud-native school operating system with **12 integrated modules**:

1. **Smart Attendance (Staff & Student)** — QR code and manual check-in/out with late detection, department and class breakdowns, real-time dashboard with auto-refresh. Configurable late cutoff and closing times.
2. **QR Code Scanner** — Native barcode detection with pure-JavaScript fallback. Camera-based scanning for staff ID cards and student IDs. Now with improved reliability, auto-start, and stale-closure fixes.
3. **Duty Roster Engine** — 8 duty types (Morning Talk, Assembly Duty, Inspection/Report, Time, Sanitary, Labour, Dining Hall, Sports). Auto-rotation algorithm, status tracking, Telegram notifications for daily assignments.
4. **AI Timetable Generator** — Constraint-satisfaction algorithm producing optimized, conflict-free timetables. Supports double periods, Friday 30-min periods, morning priority for hard subjects, teacher-subject-class-room constraints. Quality metrics dashboard.
5. **Muster Parade Management** — Digital parade session tracking, briefing delivery, task assignment with acknowledgment workflow, priority/deadline tracking, and live response feed for commandant.
6. **Prefect Roles System** — 18 prefect roles (Head Boy/Girl through Time Prefect). Dropdown assignment on student cards, prefect badges, and filter-by-prefect search. Role-based leadership tracking.
7. **Daily Intelligence Reports** — Structured reporting with activity descriptions, challenges, and suggestions. AI-generated summaries and trend analysis. Student activity reports by class teachers.
8. **Multi-Channel Communications** — Telegram bot (25+ commands including /automate, /duty, /tasks, /attendance), WhatsApp Cloud API with templates, SMS via Termii + Africa's Talking. Cascade fallback: Telegram → SMS → WhatsApp → Queue.
9. **Automation Engine** — 13 scheduled rules handling duty roster notifications, check-in reminders, absentee alerts (configurable threshold), next-period reminders, daily summary broadcasts, parade auto-close, duty auto-assign, assembly reminders, scheduled broadcast processing, and end-of-day digest. All notifications now delivered via Telegram.

10. **Global Search** — Ctrl+K or sidebar search across all 30+ platform features. Keyboard-navigable modal with section grouping.
11. **AI Assistant** — Natural language query interface with OpenAI (GPT-4o), Google Gemini, and Ollama (Qwen3:4b free local option). Database function calling for instant attendance, timetable, and staff insights.
12. **Notifications Hub** — Full notification history with sent/failed/queued status. Print queue for offline distribution. SMS queue with retry mechanism.

Technology & Architecture

Layer	Technology	Version / Detail
Frontend Framework	Next.js	16 (App Router), React 19, TypeScript
Styling	Tailwind CSS	v4, Lucide Icons
Backend	Next.js API Routes	50+ endpoints, Serverless
Database	Supabase PostgreSQL	Real-time, Row-Level Security, 30+ tables
Authentication	Supabase Auth	Password-based, role hierarchy (4 tiers)
CAPTCHA	Cloudflare Turnstile	Invisible, privacy-first
Hosting	Vercel	Global CDN, Edge Functions, Cron Jobs
AI Providers	OpenAI / Gemini / Ollama	GPT-4o, Gemini Pro, Qwen3:4b
Telegram	Bot API	Free, 25+ custom commands
WhatsApp	Meta Cloud API	v22.0, template + text messages
SMS Gateways	Termii / Africa's Talking	Nigerian + pan-African coverage
CI/CD	GitHub → Vercel	Auto-deploy on push

Security Architecture

- **Row-Level Security (RLS)** — Every database query enforces per-user access policies. Staff see only their own data.
- **Role-Based Access Control** — 4-tier hierarchy: Commandant (100) > Admin (80) > Teacher (40) > Support (20). Granular action-level permissions.
- **Service Role Isolation** — Admin API operations use a separate service-role client with elevated privileges, never exposed to client-side code.
- **Complete Audit Trail** — All overrides, staff edits, and task changes logged with user identity and timestamp.

- **Zero-Knowledge Authentication** — Supabase Auth handles all credentials. We never see or store passwords.
- **Request Validation** — Body size limits, type checking, and input sanitization on all write endpoints.

Market Opportunity

Target Market Size

Segment	Count	Annual Potential
Federal Unity Colleges (Nigeria)	115	NGN 173M
State Government Secondary Schools	> 6,000	NGN 3.6B
Private Secondary Schools (Nigeria)	> 15,000	NGN 9B
Pan-African Schools (Ghana, Kenya, SA)	~30,000	NGN 18B

At a conservative NGN 600,000/school/year license fee.

Competitive Advantage

- **Only platform combining operations management + academic scheduling + communications + automation** — No competitor offers this full stack.
- **Built by educators, for educators** — Developed at AFCS, refined through daily production use by actual school administrators. Every feature is user-driven.
- **Telegram-native communications** — 25+ bot commands running on zero-cost infrastructure. Works on 2G networks with basic phones. No app installation required.
- **Offline-resilient architecture** — Serverless edge deployment with Supabase real-time sync, SMS fallback, and notification queue for retry.
- **Lowest total cost of ownership** — Serverless infrastructure (Vercel + Supabase) keeps hosting costs near zero even at scale.
- **Nigerian-built, NDPR-compliant** — Understands local school culture, complies with Nigeria Data Protection Regulation.
- **Production-proven** — Live since early 2026 at AFCS Igbara-Oke, processing hundreds of daily transactions.

Business Model

Pricing Tiers

Tier	Features	Annual Fee
Essential	Attendance + Duty Roster + Reports	NGN 350,000
Professional	All modules + AI Timetable + Telegram Bot + Automation	NGN 600,000
Enterprise	All modules + WhatsApp/SMS + Dedicated Support + SLA	NGN 1,200,000

Additional Revenue Streams

- **Deployment & Training** — NGN 250,000 one-time setup fee per school (onboarding, data migration, staff training)
- **Hardware Bundles** — QR scanners, tablets, biometric devices (10-15% margin)
- **Annual Maintenance** — 15% of license fee for priority support and updates
- **SMS Credits** — Volume-based SMS pricing (buy bulk, resell per message)
- **Data Analytics** — Premium reporting dashboard for education boards/ministries
- **Custom Integration** — API access, custom dashboard, white-labeling for school groups

Financial Projections

Metric	Year 1	Year 2	Year 3	Year 5
Schools Deployed	10	35	80	200
Avg Revenue per School	NGN 750K	NGN 700K	NGN 650K	NGN 600K
Annual Recurring Revenue	NGN 7.5M	NGN 24.5M	NGN 52M	NGN 120M
Setup Fees (one-time)	NGN 2.5M	NGN 6.25M	NGN 11.25M	NGN 3.75M
Gross Revenue	NGN 10M	NGN 30.75M	NGN 63.25M	NGN 123.75M

Operating Costs	NGN 8M	NGN 15M	NGN 25M	NGN 40M
Net Profit	NGN 2M	NGN 15.75M	NGN 38.25M	NGN 83.75M
Margin	20%	51%	60%	68%

Product Maturity & Milestones

Current Feature Status (July 2026)

Feature	Status	Notes
Staff Attendance (QR + Manual)	Production	Check-in/out, late detection, duration tracking, daily override
Student Attendance (QR + Manual)	Production	3-period tracking (morning/afternoon/evening), class-based
Duty Roster Engine	Production	8 duty types, auto-rotation, generation, day/week views
AI Timetable Generator	Production	Constraint satisfaction, conflict detection, quality metrics
Muster Parade Management	Production	Sessions, briefings, tasks with acknowledge/complete flow
Prefect Roles System	Production	18 roles, student assignment, badges, filters
Telegram Bot (25+ commands)	Production	Attendance, duties, tasks, automations, broadcasts
WhatsApp Cloud API	Production	Template + text messages, task notifications
Automation Engine (13 rules)	Production	Scheduled + event-driven, Vercel Cron integration
Global Search	Production	Ctrl+K, 30+ features, keyboard navigation
AI Assistant	Production	OpenAI + Gemini + Ollama, database function calling
QR Scanner	Production	BarcodeDetector + jsQR, 7 reliability fixes applied
Notifications Hub	Production	History, queue, print, status dashboard
Daily Reports + AI Summaries	Production	Structured reporting, trend analysis
SMS (Termii + Africa's Talking)	Production	Fallback channel with queue + retry
Staff & Student CRUD	Production	Full create/read/update/delete with role-based access
System Settings & Config	Production	Attendance rules, school info, prompts, templates

Development Phases Completed

- **Phase 1 (100%):** Staff Attendance — QR/manual check-in/out, late detection, real-time dashboard
- **Phase 2 (100%):** Student Attendance — Class-based tracking, 3-period system, teacher attendance
- **Phase 3 (100%):** Duty Roster & Reporting — 8 duty types, auto-assignment, daily reports, AI summaries
- **Phase 4 (100%):** Muster Parade Automation — Sessions, briefings, tasks, acknowledgments
- **Phase 5 (100%):** AI Timetable Generator — Constraint-satisfaction algorithm, quality metrics, history
- **Phase 6 (100%):** Multi-Channel Communications — Telegram bot, WhatsApp API, SMS fallback, automation engine
- **Phase 7 (100%):** Prefect Roles & Global Search — Student leadership tracking, cross-feature search

Use of Funds

Category	Allocation	Amount	Details
Product Development	40%	NGN 20M	Multi-tenancy architecture, white-labeling, mobile apps (React Native), analytics dashboard, e-exam platform
Sales & Marketing	30%	NGN 15M	Sales team (3 FTE), school visitation program, demo events at AFCS commands, digital marketing, conference attendance
Operations & Support	20%	NGN 10M	24/7 support desk, deployment engineers, training material production, self-paced onboarding portal
Infrastructure & Compliance	10%	NGN 5M	Cloud costs (Vercel + Supabase), penetration testing, SOC 2 readiness, NDPR compliance audit

Risk Mitigation

Risk	Probability	Impact	Mitigation
Slow school adoption	Medium	High	Free pilot program for first 10 schools; testimonial-driven sales; ROI calculator for school leadership
Internet reliability	High	Medium	Telegram bot works on 2G; offline check-in queue; SMS fallback; cached dashboard data
Competition	Medium	Medium	Unique ops-management focus; first-mover in this niche; 18-month head start
Staff turnover at client schools	High	Low	Self-paced training portal; admin delegation built into product; no single point of failure
Data security breach	Low	Critical	Supabase RLS + service-role isolation; regular security audits; no password storage
Regulatory changes	Low	Medium	NDPR-compliant architecture; data residency in Nigeria (via Supabase); adaptable to new regulations

Founder dependency	Medium	High	Comprehensive documentation; code quality standards; team hiring plan in Year 1
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Investment Ask

Investment Amount: NGN 50,000,000 (~\$32,000 USD)

Structure: Seed Round — 15% Equity

Use of Funds: Product development (multi-tenancy, mobile apps), sales team expansion, 10-school pilot deployment

Timeline: 24 months to deploy across 100 Nigerian schools

ROI for Schools: 10x reduction in admin time, 100% audit trail, real-time operational visibility, zero communication cost via Telegram

ROI for Investors: Path to NGN 120M ARR by Year 5 with 68% net margins. Scalable to NGN 1B+ across Africa.

Why Now?

- **Proven product, real traction** — Live at AFCS with 80+ staff and 3,000+ students since early 2026. All 12 core modules in production.
- **Growing market** — Nigerian edtech market projected to reach \$3.1B by 2028 (GSMA). Federal Ministry of Education driving digital transformation.
- **Mobile-first, low-barrier adoption** — 85%+ of Nigerian teachers use WhatsApp/Telegram daily. Zero app installation required. Works on \$30 phones.
- **First-mover advantage** — No competitor offers integrated operations management + academic scheduling + automation for African schools.
- **Scalable architecture** — Serverless stack scales from 50 to 50,000 users without re-architecture. Vercel + Supabase handle global scale.
- **Government tailwind** — Federal Ministry of Education Smart Education initiative; AFCS commands across 6 geo-political zones as natural expansion path.

- **AI cost collapse** — Ollama + Qwen3:4b provide free, private AI inference. OpenAI costs continue to drop. Our AI features become cheaper over time.

Roadmap: Post-Investment

Quarter	Milestone	Deliverable
Q3 2026	Multi-Tenancy & White-Label	Separate school instances, custom branding, per-school admin panel
Q3 2026	Mobile App (React Native)	Staff check-in/out, task management, notifications on mobile
Q4 2026	Parent Portal	Real-time absence alerts, report cards, fee payment tracking, behavior reports
Q4 2026	E-Examination Platform	Secure digital exams with auto-grading, anti-cheating, result analytics
Q1 2027	Asset & Inventory Management	QR-code asset tracking, textbook distribution, lab equipment, maintenance alerts
Q1 2027	AI Counselling & Welfare	Flags attendance drops, academic decline, behavioral patterns for guidance counselors
Q2 2027	10-School Pilot Deployment	AFCS commands in 3 geo-political zones, 5 private schools
Q3 2027	Business Intelligence Suite	Cross-school analytics for education boards, trend dashboards, performance benchmarking
Q4 2027	Pan-African Expansion	Ghana, Kenya, South Africa — local currency pricing, regional hosting

The Team

- **Built by educators and technologists** at Air Force Comprehensive School, Igbara-Oke — the team that uses the product daily.
- **Deep domain expertise** — Intimate understanding of boarding school operations, military command structure, duty culture, and the unique challenges of Nigerian secondary education.
- **Technical excellence** — Modern full-stack JavaScript (Next.js 16, TypeScript, Tailwind CSS v4), cloud infrastructure (Vercel, Supabase), AI integration (OpenAI, Gemini, Ollama), multi-channel messaging (Telegram API, Meta Cloud API, SMS gateways).
- **User-driven development** — Every feature requested, tested, and validated by actual school administrators and commandant. The product roadmap is dictated by real user needs, not assumptions.
- **Lean, capital-efficient** — Built to production with minimal external funding. Proves the team's ability to execute efficiently.

Contact

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